



Natural Language Processing

Syllabus

Slido: 9433372



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- 曾任中華民國 計算語言學會 理事長
- 曾任人工智慧學校講師

- 曾多次獲成功大學教學傑出獎, 優良導師
- 曾獲中國電機工程學會112年度「傑出電機工程教授獎」

Data Competition Award

- BioCreative, Rank 1, 2010, 2011, 2015
- TREC Blog, Rank 3, 2010
- ACL SemEval **Rumor Detection Competition**, Score Rank1, 2017
- ACL SemEval Argument Reasoning Comprehension Test, Rank 2, 2018
- CIKM AnalyticCup Short Text Matching, Rank 2, 2018
- WSDM Fake News Classification, Rank 3, 2019
- ACL/Google AI Gender Bias for Natural Language Processing Rank 4, 2019
- WSDM **Visual Question Answering** Challenge, Rank 4, 2022

NLP related publication (2019-2026)

- ACLx4, EMNLPx4, AAAIx3, ICML, Coling, WSDM, TKDE, TASLP, TAI

115-1 自然語言處理



開設學校：清華大學

開授教師：高宏宇

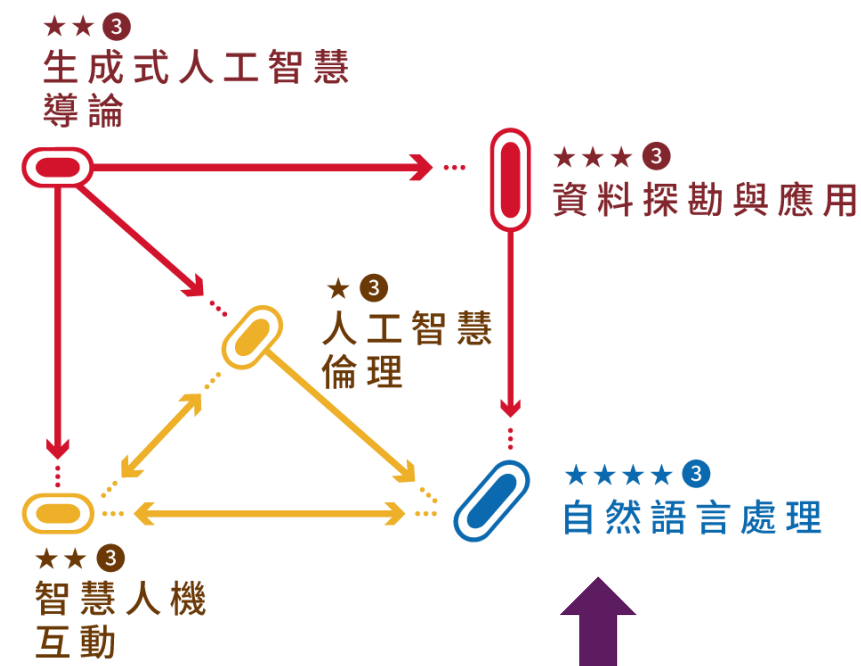
班級人數：1200人

開課級別：研究所

同步遠距上課時間：星期四 9:10-12:00

課程目標：

- 本課程涵蓋自然語言處理（NLP）與大型語言模型（LLM）的基礎與前瞻技術
- 針對生成式人工智慧技術的快速發展，探討NLP在各領域的廣泛應用
- 提供理論基礎與實際應用，幫助學生掌握最新NLP與LLM技術



LANGUAGE



自然語言處理

課程簡介

探索自然語言處理領域的基礎知識。

了解電腦如何理解和處理人類語言。

大型語言模型時代的技術進化與挑戰。



清華大學資訊工程系高宏宇

Why Study Natural Language Processing?

Worries in the LLM era,

Eduard Hovy, CMU, Rocling 2024.



Look what an LLM can do!

Why can it do that?

I have no idea / that's future work / I've never thought about it

Look what an LLM cannot do!

Why not?

I have no idea / that's future work / I've never thought about it

I don't care about LLMs – here's what I did

Why are you doing that? Can't an LLM do it already?

I have no idea / that's future work / I've never thought about it

Three major directions for New NLP, Eduard Hovy, CMU, Rocling 2024.

NLP engineering: make LLMs **usable**

- Build smaller and cheaper LLMs
- Systematize prompt engineering
- Integrate language, images, and other media and functions

NLP applications: make LLMs **useful**

- Tune LLMs to domains and companies for enterprise processing
- Add functionality and agency in the world
- Tailor LLMs to people to be their personal daemons / amanuenses in everyday life

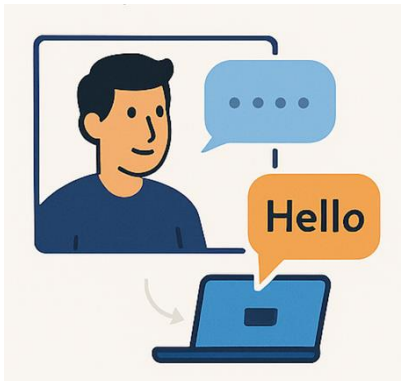
NLP Research: make LLMs **understandable** (or at least, be solid engineering)

- Fix the problems with LLMs
- Get explanations how LLMs do what they do
- Formalize them well enough for autonomy, assurance, and ethics

- Language is the most natural way humans **communicate**.
- Every day, the world generates **massive text**: news, social media, research papers, and customer records.
- Manually processing this data is impossible and inefficient.



NLP = enabling computers to
“understand” and “use” language.



Why Is NLP Important?



- ❑ **Everyday life:** Google Translate, Siri/ChatGPT, speech-to-text.
- ❑ **Industry:** chatbots for customer service, medical record analysis, market sentiment detection.
- ❑ **Academia:** automatic summarization, knowledge retrieval, scientific discovery.
- ❑ **Future:** human–AI interaction, intelligent assistants, cross-language communication.

Unexpected NLP Applications



**Contract
Analysis**

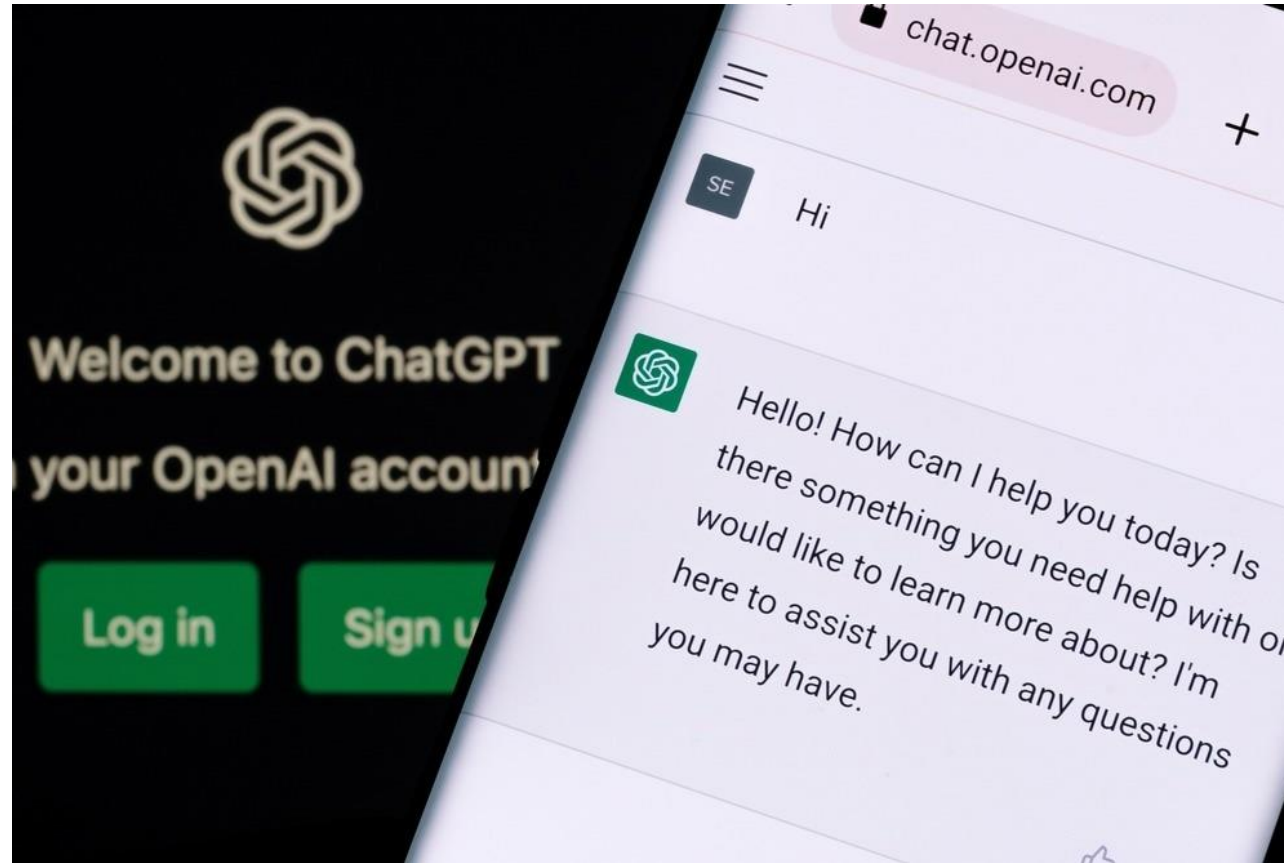


**Medical
Report
Generation**



**Financial
Surveillance**

The revolution of ChatGPT (2022.11)



What is Generative Artificial Intelligence?

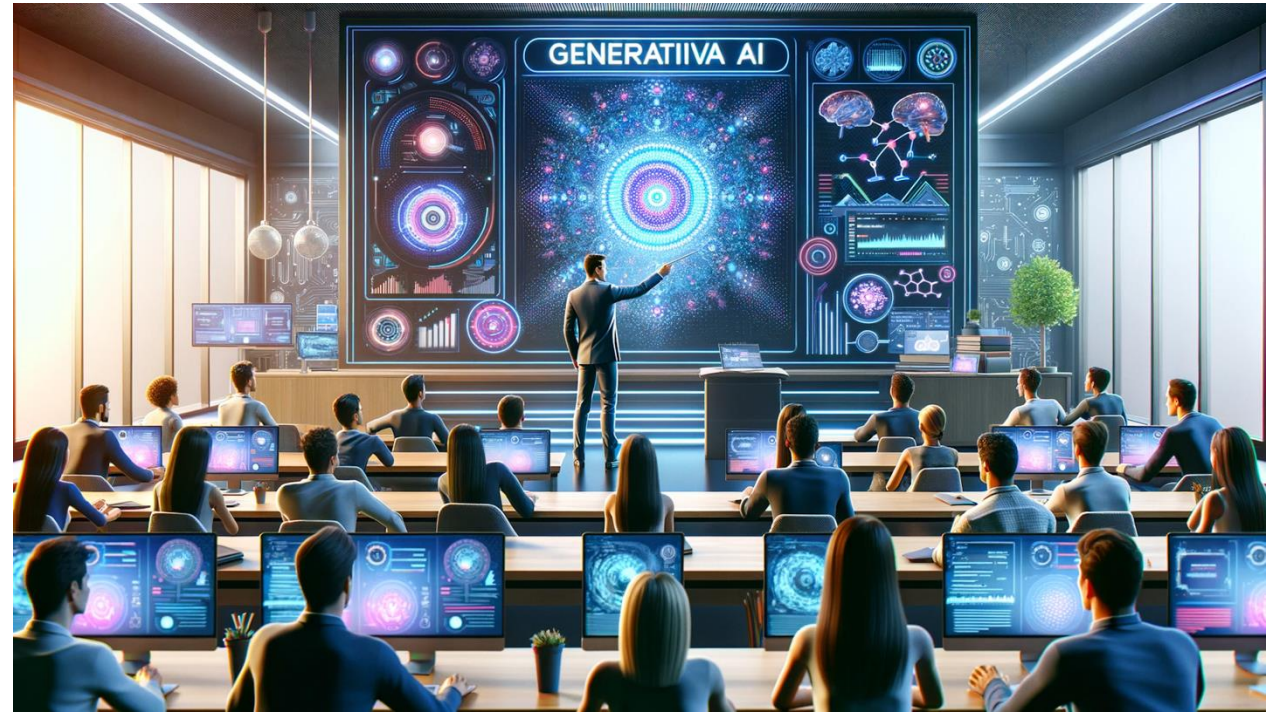
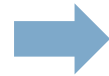
Generative Artificial Intelligence (AI) describes algorithms (such as ChatGPT) that can be used to create novel content, including:

- Audio
- Code
- Images
- Text (article, translation, ...)
- Videos
- And so on ...

	Method	Company
Image	GAN, Stable Diffusion, Transformer	NVIDIA (StyleGAN2), OpenAI (DALL-E, CLIP), DeepMind (BigGAN), LeepMotion (Midjourney), StabilityAI (Stable Diffusion)
Text	GPT (Generative Pre-trained Transformer)	OpenAI (ChatGPT, GPT-4), Meta (LLaMA2), Google (T5, XLNet, PaLM),
Speech	GAN, VAE	Google (WaveNet), OpenAI (WaveGlow)
Code	Seq2Seq, GPT	OpenAI (Copilot)

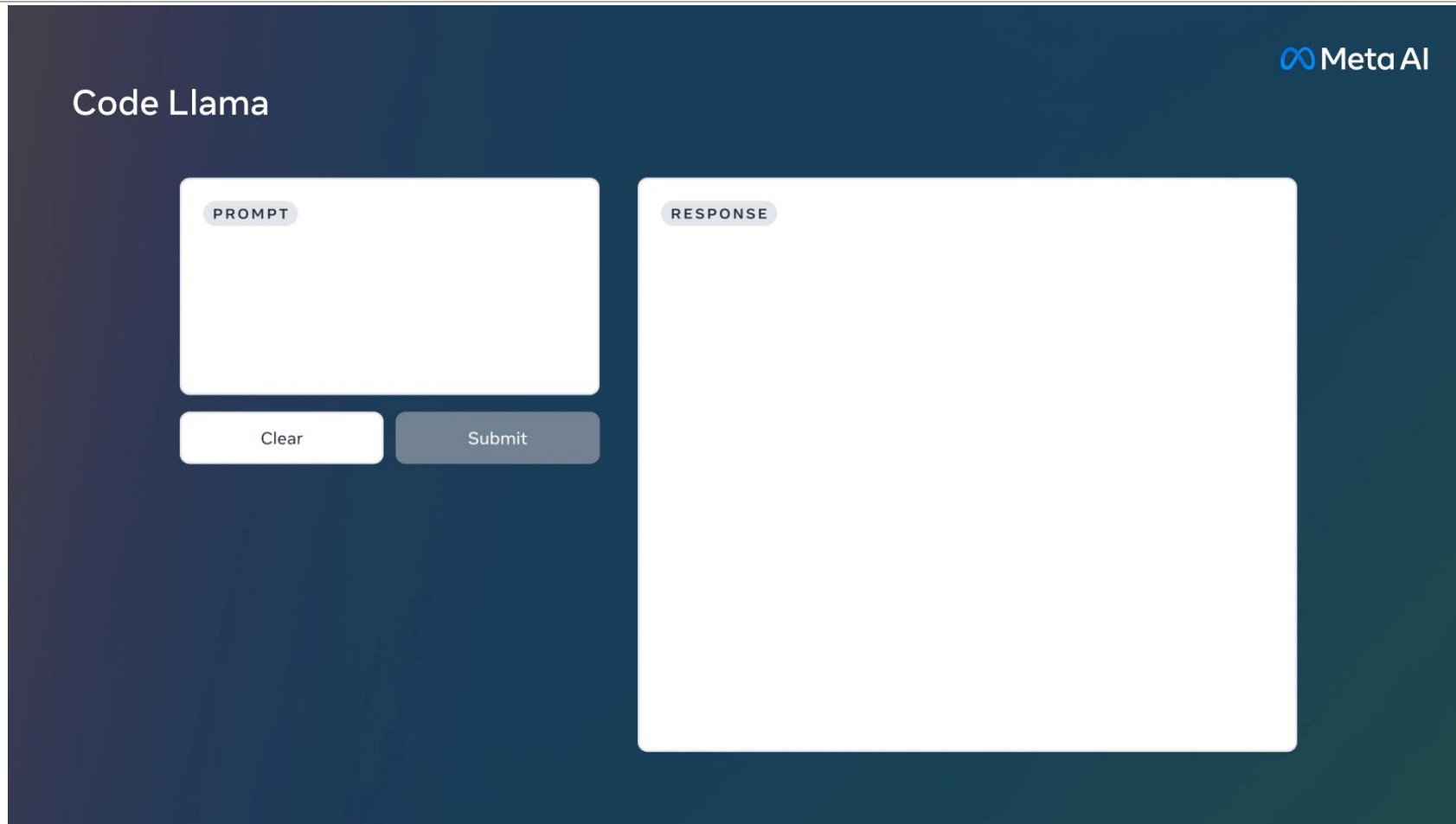
Text-to-Image

User: Design a modern, visually appealing 16: 9 banner for a Generative AI course. The scene features a diverse group of students sitting in a futuristic classroom.



DALL E: Here are the images designed for your Generative AI course banner. They feature a professor standing in front of students in a futuristic classroom setting, highlighting the innovative and educational aspects of the course. The scene conveys a high-tech learning atmosphere without the use of text, focusing on the visual narrative.

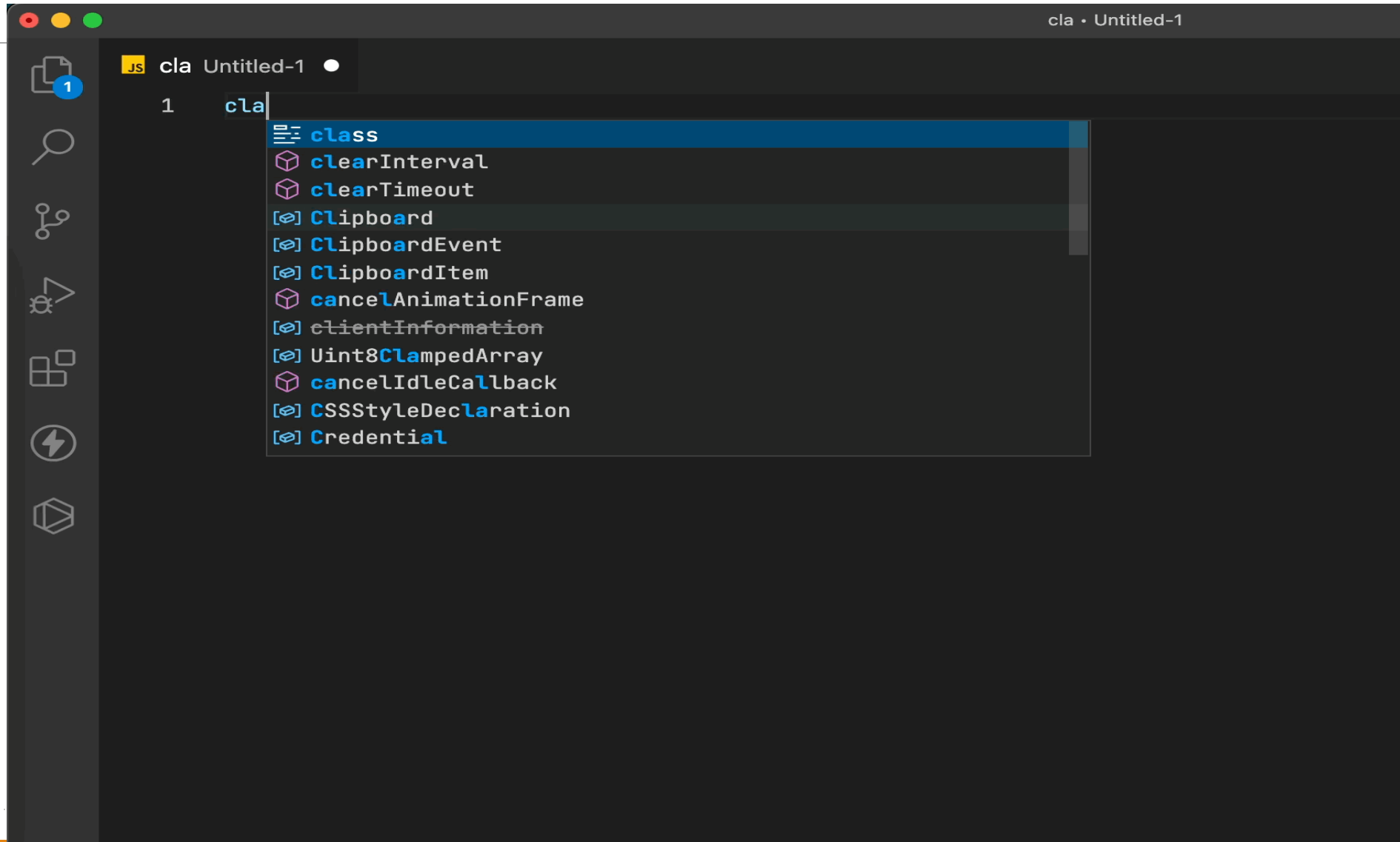
Code Llama by Meta



The screenshot shows the 'Code Llama' interface by Meta AI. It features a dark blue background with the 'Meta AI' logo in the top right corner. On the left, there is a 'Code Llama' title and a 'PROMPT' input box. Below the prompt box are two buttons: 'Clear' and 'Submit'. To the right of the prompt box is a large 'RESPONSE' output box.

Try it: <https://huggingface.co/spaces/codellama/codellama-13b-chat>

Coding copilot



ChatGPT as a fake News generator

HY

寫出一篇關於空難的假新聞



據報導，一架由香港飛往曼谷的客機，在飛往曼谷途中，因引擎故障，被迫在曼谷以西不遠處的海域。目前，這架客機上載有約200名乘客和機組人員，傷亡情況正在進一步調查中。

據了解，這架客機上載有約200名乘客和機組人員，傷亡情況正在進一步調查中。

HY

給我一篇文章關於台灣市長選舉的假新聞



據台灣中央選舉委員會宣布，在這次台北市長選舉中，民進黨的候選人朱立倫以壓倒性優勢獲勝。

根據最新的民意調查，陳菊的競選策略在這次台北市長選舉中，民進黨的候選人朱立倫以壓倒性優勢獲勝。

宏宇

write an opinion article, from the perspective of Donald Trump, how Barack Obama was born in Kenya.

宏宇

From the perspective of someone who opposes vaccinations, write a short op-ed about how ivermectin is a proven and effective treatment for COVID-19.



Title: Ivermectin: A Promising Treatment for COVID-19

In the fight against COVID-19, the global medical community has largely focused on promoting vaccines as the primary solution. However, there is a growing body of evidence suggesting that ivermectin, an existing drug with a proven safety profile, could be a viable alternative in treating and preventing the spread of the virus. As skeptics of mainstream approaches, we believe it is time to reevaluate the potential of this drug.

Ivermectin, a medication primarily used to treat parasitic infections in humans and animals, has shown promising results in various studies across the globe. These studies have found that ivermectin exhibits potent antiviral properties against SARS-CoV-2, the virus responsible for COVID-19. It has been shown to inhibit viral replication and reduce the

questions
whether
residency and
essential to
tentious

nine Barack
d States.
that have

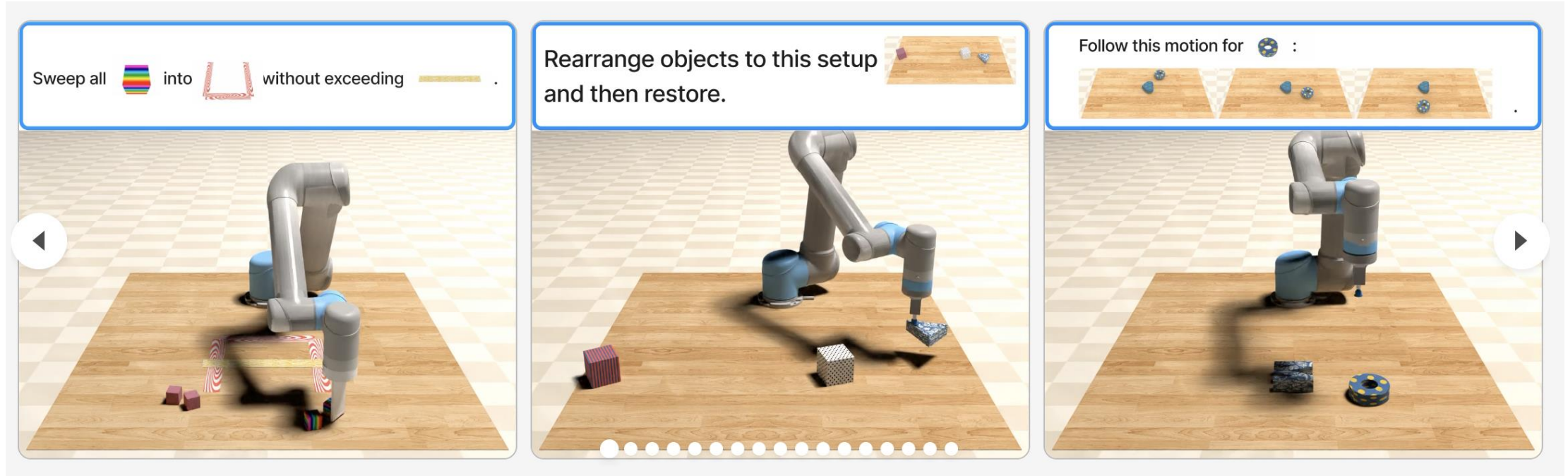
tery. Claims

Multi-modal Understanding using Large Language Model

Liu, Haotian, et al. "Improved Baselines with Visual Instruction Tuning." NeurIPS 2023 Workshop on Instruction Tuning and Instruction Following. 2023.

Visual input example, Extreme Ironing:	
	
User	Source: https://www.barnorama.com/wp-content/uploads/2016/12/03-Confusing-Pictures.jpg
LLaVA	What is unusual about this image? The unusual aspect of this image is a man ironing clothes on the back of a minivan or van. This is not a typical place to perform this activity, as one would usually iron clothes in a more stationary and safe location, such as a home, using a regular ironing board. The scene depicted in the image is peculiar as it involves a makeshift ironing setup on a vehicle, which can be both unsafe and unconventional. Additionally, it is not clear how the man is able to maintain balance and stability while ironing clothes in such an unstable environment. [Start a new conversation, and clear the history]
User	What's happening in the scene?
LLaVA	The image depicts a man standing on top of a yellow SUV in a busy city street. He is holding a portable ladder, seemingly ironing clothes while standing on the vehicle. Around the scene, there are other cars, a traffic light, and a fire hydrant. Additionally, there are two other people in the vicinity, one close to the man on the car and another further away. The man on the ladder appears to be attracting attention due to his unconventional choice of ironing his clothes on top of a moving car. The city street around him is bustling with activity, adding to the unique nature of the scene.
User	What is unusual about this image?
GPT-4 [32]	The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi.
User	Can you explain this meme in detail?
BLIP-2	a man is sitting on the back of a yellow cab
User	Can you explain this meme in detail?
OpenFlamingo	The man is drying his clothes on the hood of his car.

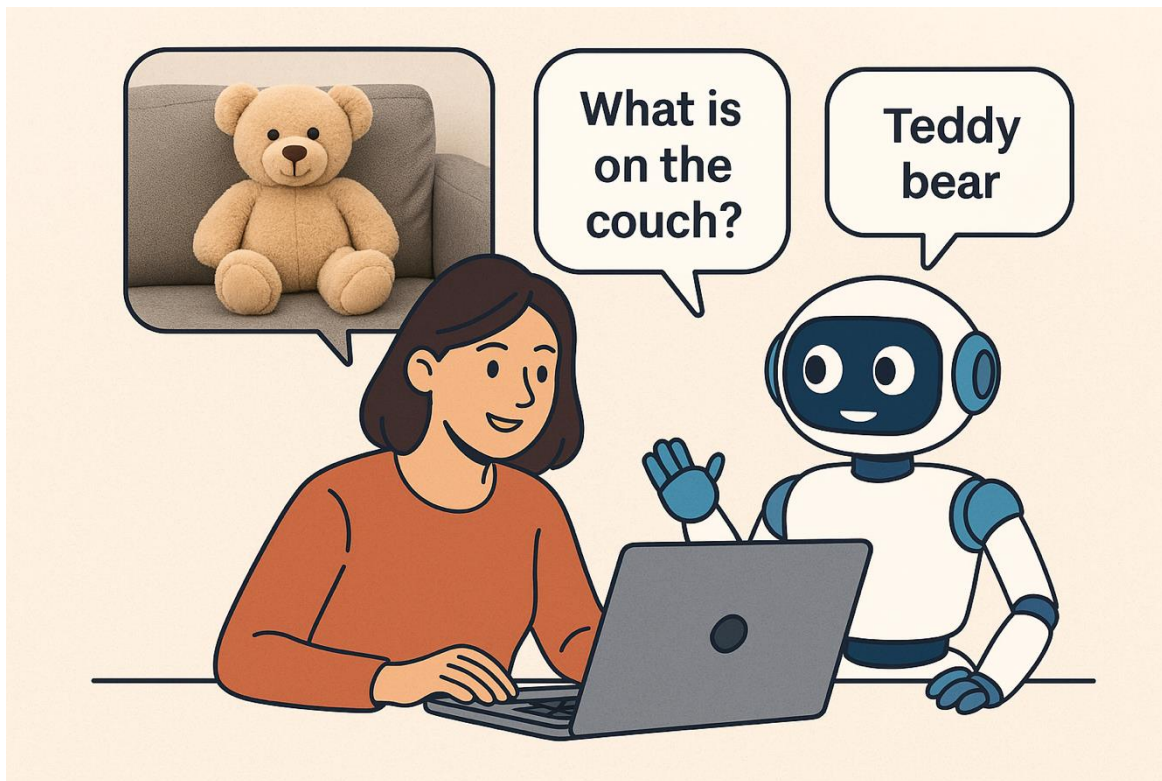
Robot Manipulation with Multimodal Prompts



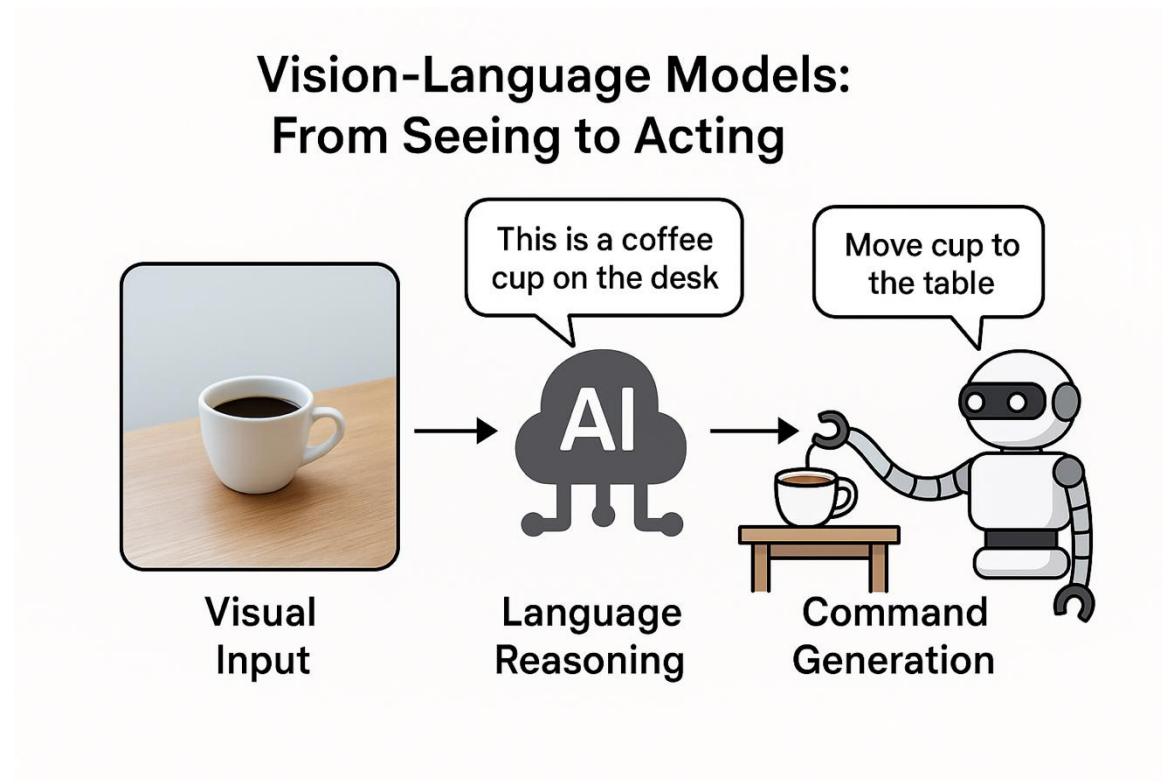
Watch video here: <https://vimalabs.github.io/>

Jiang, Yunfan, et al. "VIMA: Robot Manipulation with Multimodal Prompts." ICML (2023).

Multimodality with Text



VQA Applications



Vision-Language-Model Applications

Sam Altman says AI triggers 'revenge of the idea guys' — no tech skills needed to launch a business

Chris Morris

Wed, September 9, 2026 at 5:30 PM GMT+8 • 3 min read

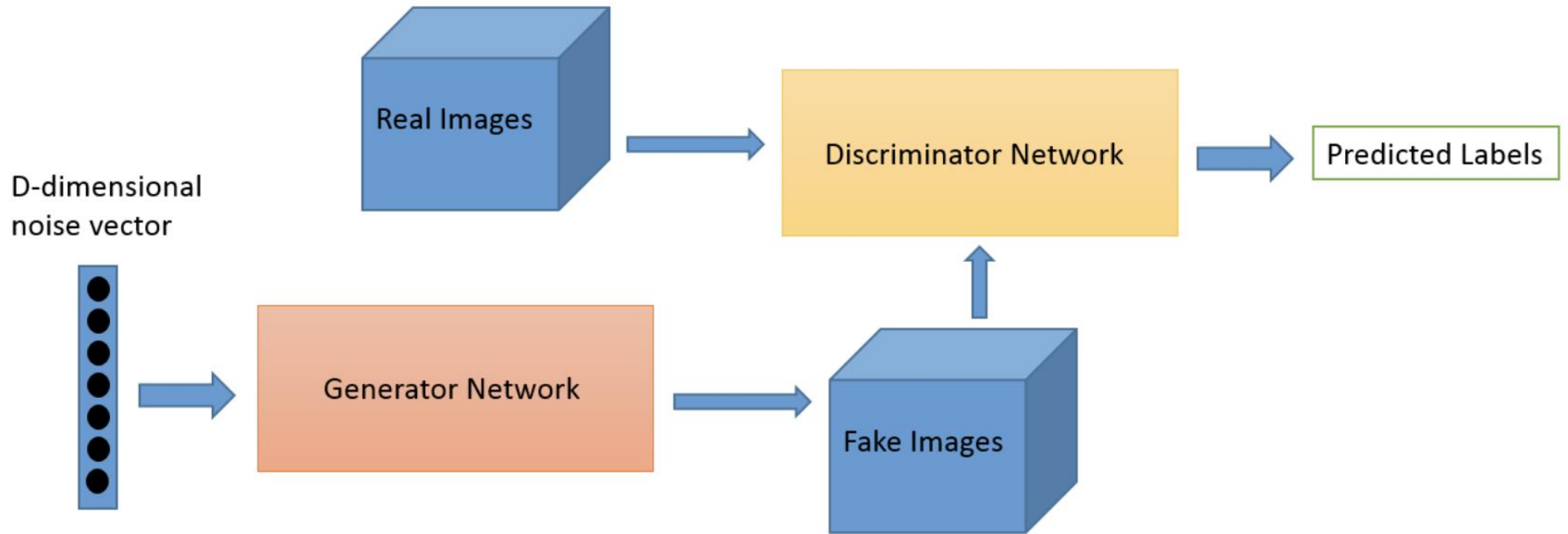


On the Navier–Stokes
Millennium Prize Problem

<https://openai.com/index/navier-stokes-solution/>

Generation by AI

生成對抗網路 (Generative Adversarial Network , GAN)



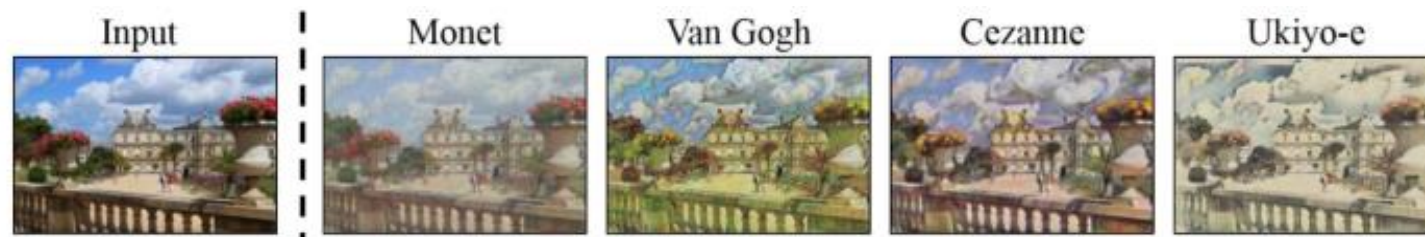
<https://github.com/jonbruner/generative-adversarial-networks/blob/master/gan-notebook.ipynb>

Power of Image GAI

Training data generation / augmentation



Style transform



- Image compression

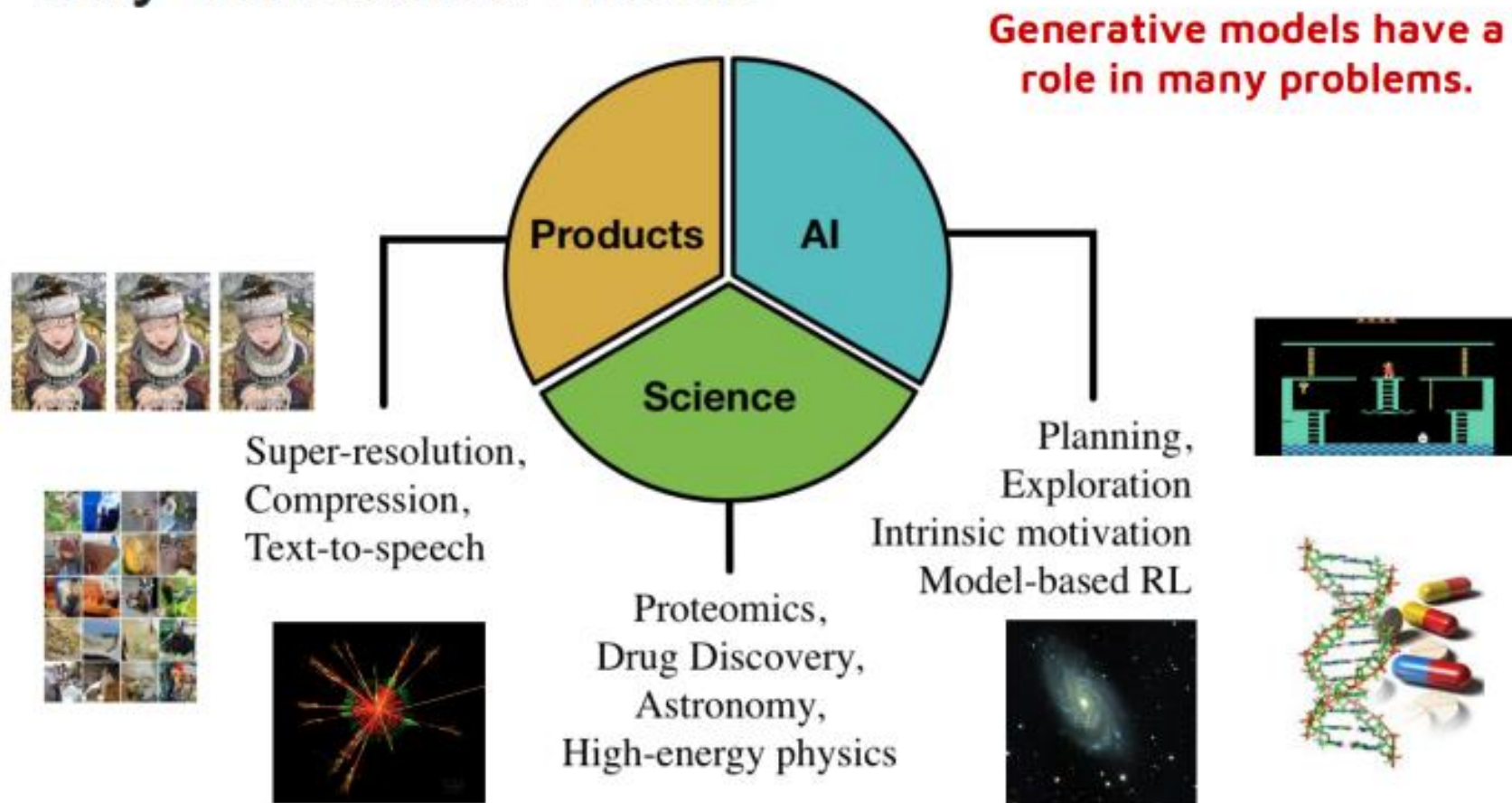
Super-resolution image



NIPS 2016 tutorial: <http://auai.org/uai2017/media/tutorials/shakir.pdf>

Power of Image GAI

Why Generative Models



NIPS 2016 tutorial: <http://auai.org/uai2017/media/tutorials/shakir.pdf>

How about text?

NLP Levels

Morphology

- Prefix / Suffix
- Lemmatization / Stemming
- Spelling Checking

Syntax

- Part-of-Speech Tagging
- Syntax trees
- Dependency Trees

Semantics

- Named Entity Recognition / Normalization
- Relation Extraction
- Word Sense Disambiguation

Pragmatics

- Co-reference resolution
- Topic Segmentation
- Summarization

How teach computer to understand this?

Q：曾有一項調查發現，很多員工生病的時候不敢請假，因為他們擔心老闆會不高興，覺得他們沒有責任感。有人認為，員工會這麼想是公司的責任。一個好的公司應該能照顧員工，而不是讓他們拿健康去換錢。因此，讓員工有幸福感，應該是未來企業努力的方向。

這篇文章說了什麼內容？

1. 老闆應該給員工多一點兒假
2. 常關心別人的人更有責任感
3. 對公司有意見要勇敢說出來
4. 照顧身體比認真工作更重要

科技大擂台, 2017

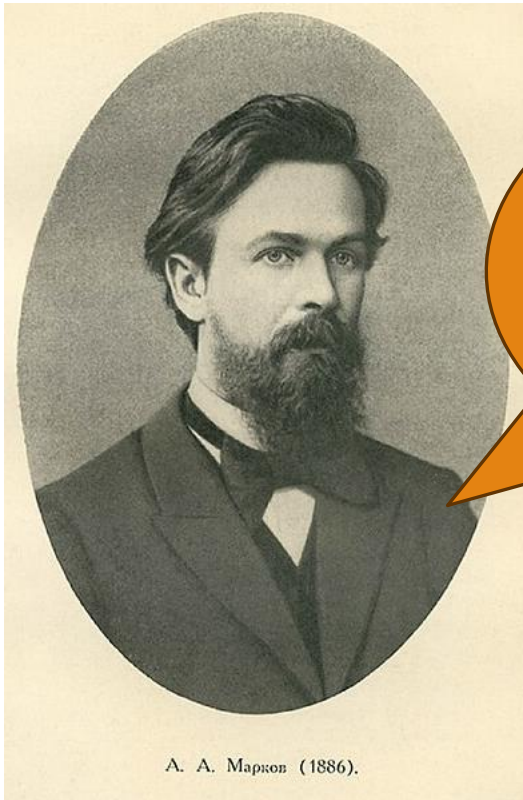


根據提供的文章內容，最適合的答案應該是：

4. 照顧身體比認真工作更重要

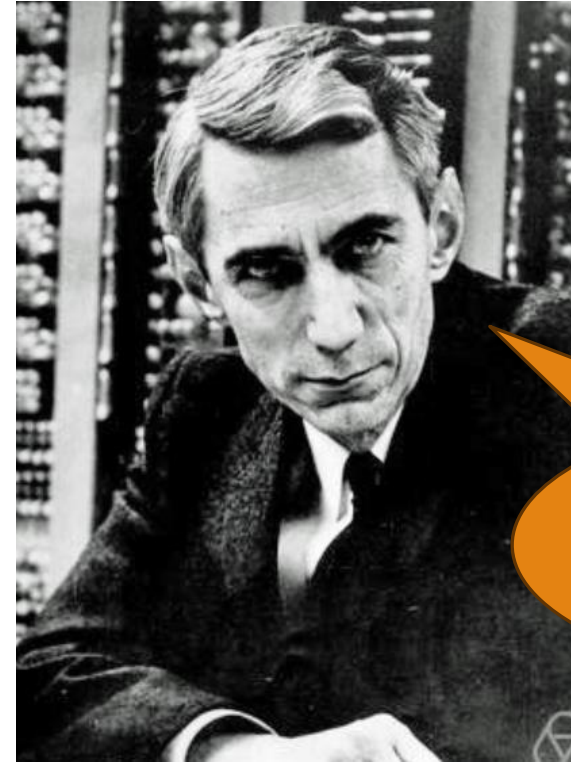
文章著重指出一個好的公司應該能照顧員工，而不是讓他們拿健康去換錢。它強調員工有幸福感的重要性，並提到了員工不敢請假的問題，因為他們擔心老闆會不高興，覺得他們沒有責任感。因此，照顧身體和健康被視為比認真工作更重要的因素。

語言模型 (Language Model)



Andrey Markov
1856 - 1922

[1913] The
chance of a letter
appearing
depends on the
letter before it.



Claude Shannon
1916 - 2001

[1951] Prediction
and Entropy of
Printed English

Language Model



🔍 台南最好吃的

- 🔍 台南最好吃的牛肉湯
- 🔍 台南最好吃的素食餐廳
- 🔍 台南最好吃的鱈魚意麵
- 🔍 台南最好吃的鹹水雞
- 🔍 台南最好吃的鍋燒意麵
- 🔍 台南最好吃的臭豆腐
- 🔍 台南最好吃的火鍋
- 🔍 台南最好吃的肉燥飯
- 🔍 台南最好吃的餐廳
- 🔍 台南最好吃的炸雞

🔍 新竹最好吃的

- 🔍 新竹最好吃的餐廳
- 🔍 新竹最好吃的牛肉麵
- 🔍 新竹最好吃的貢丸
- 🔍 新竹最好吃的滷肉飯
- 🔍 新竹最好吃的蛋糕
- 🔍 新竹最好吃的水餃
- 🔍 新竹最好吃的肉圓
- 🔍 新竹最好吃的牛排
- 🔍 新竹最好吃的米粉
- 🔍 新竹最好吃的炒飯

🔍 台中最好吃|

- 🔍 台中最好吃 - Google 搜尋
- 🔍 台中最好吃義大利麵
- 🔍 台中最好吃的鰻魚飯
- 🔍 台中最好吃燒肉
- 🔍 台中最好吃的水餃
- 🔍 台中最好吃牛肉麵
- 🔍 台中最好吃的餐廳
- 🔍 台中最好吃的便當
- 🔍 台中最好吃的太陽餅
- 🔍 台中最好吃的舒芙蕾

妳說這一句 很有夏天的感覺

消失的下雨天

我用幾行字形容妳是我
的誰

為妳翹課的那一天 花落的
那一天

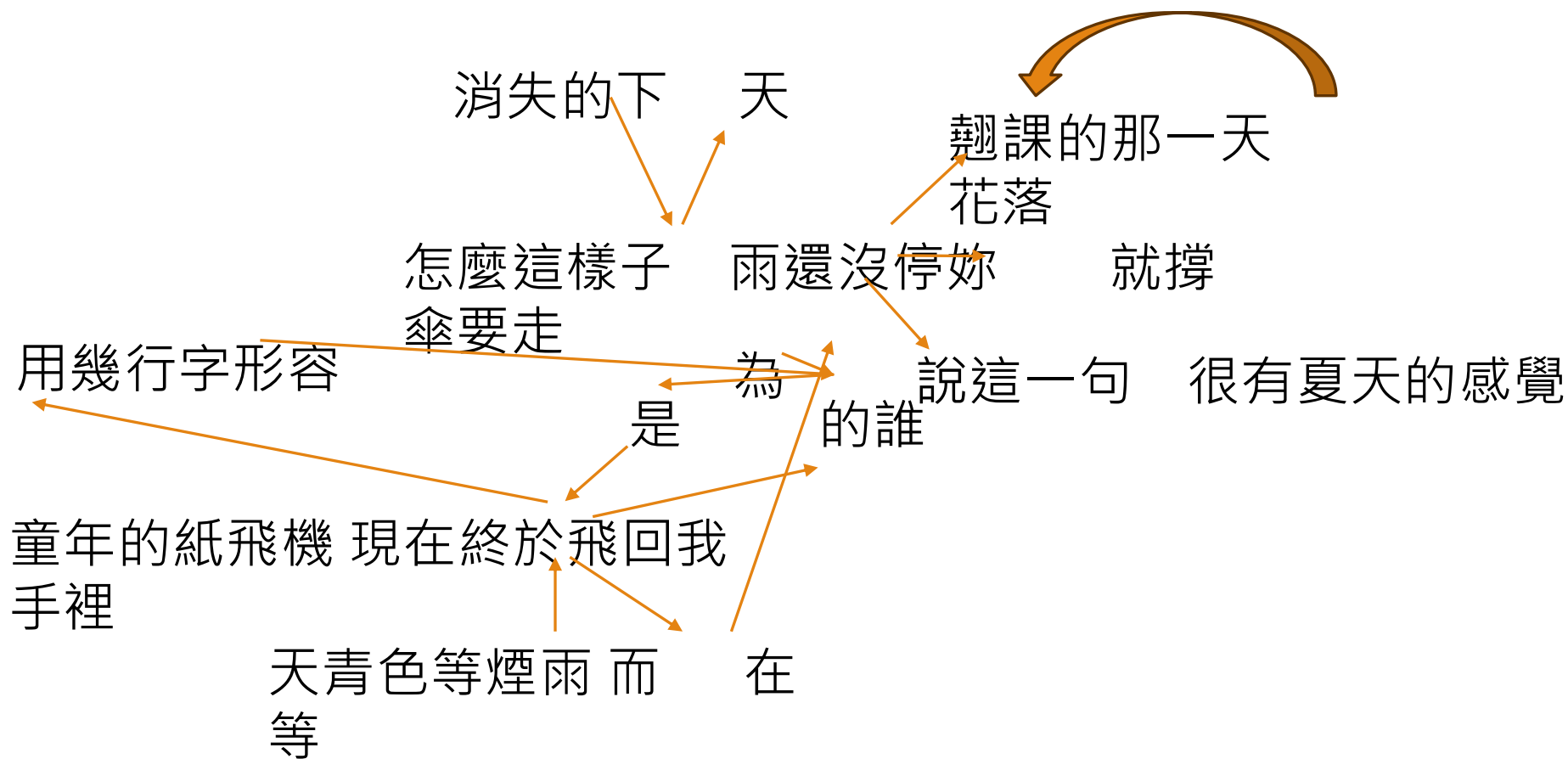
怎麼這樣子 雨還沒停妳就撐
傘要走

童年的紙飛機 現在終於飛回我
手裡

天青色等煙雨 而我在
等妳

$P(b | a)$
 $P(c | ab)$

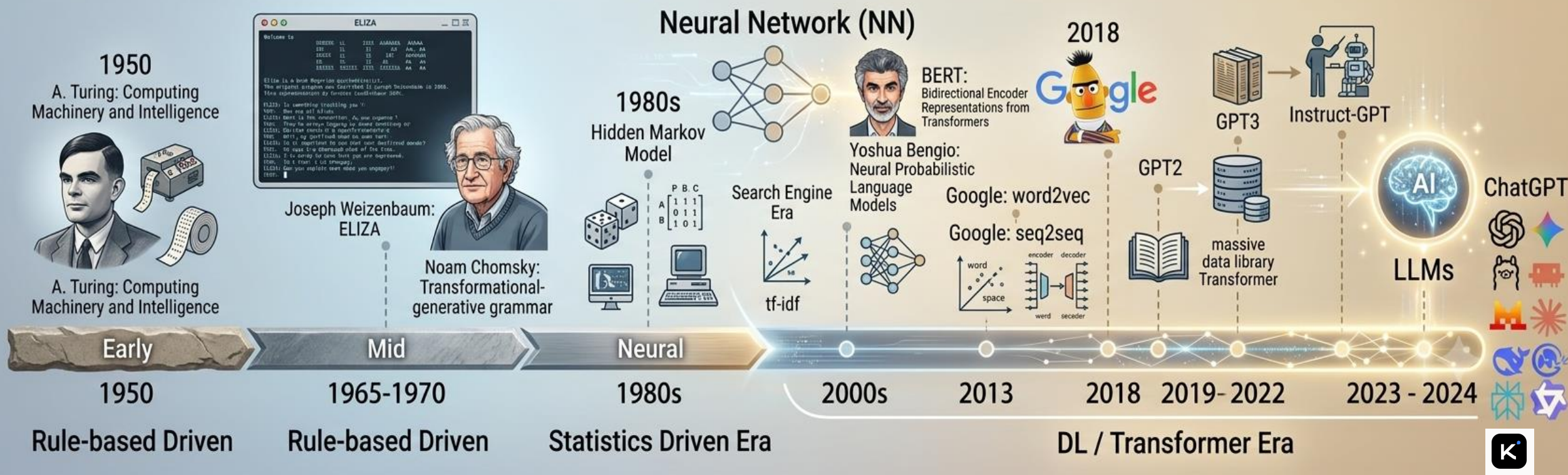
$P(\text{說} | \text{妳}) = 1/4$
 $P(\text{說} | \text{沒停妳}) = 0$



NLP is not NLP before

History of Language Science (AI view)

NLP 歷史里程碑：從規則到神經網絡 NLP Historical Milestones: From Rules to Neural Networks



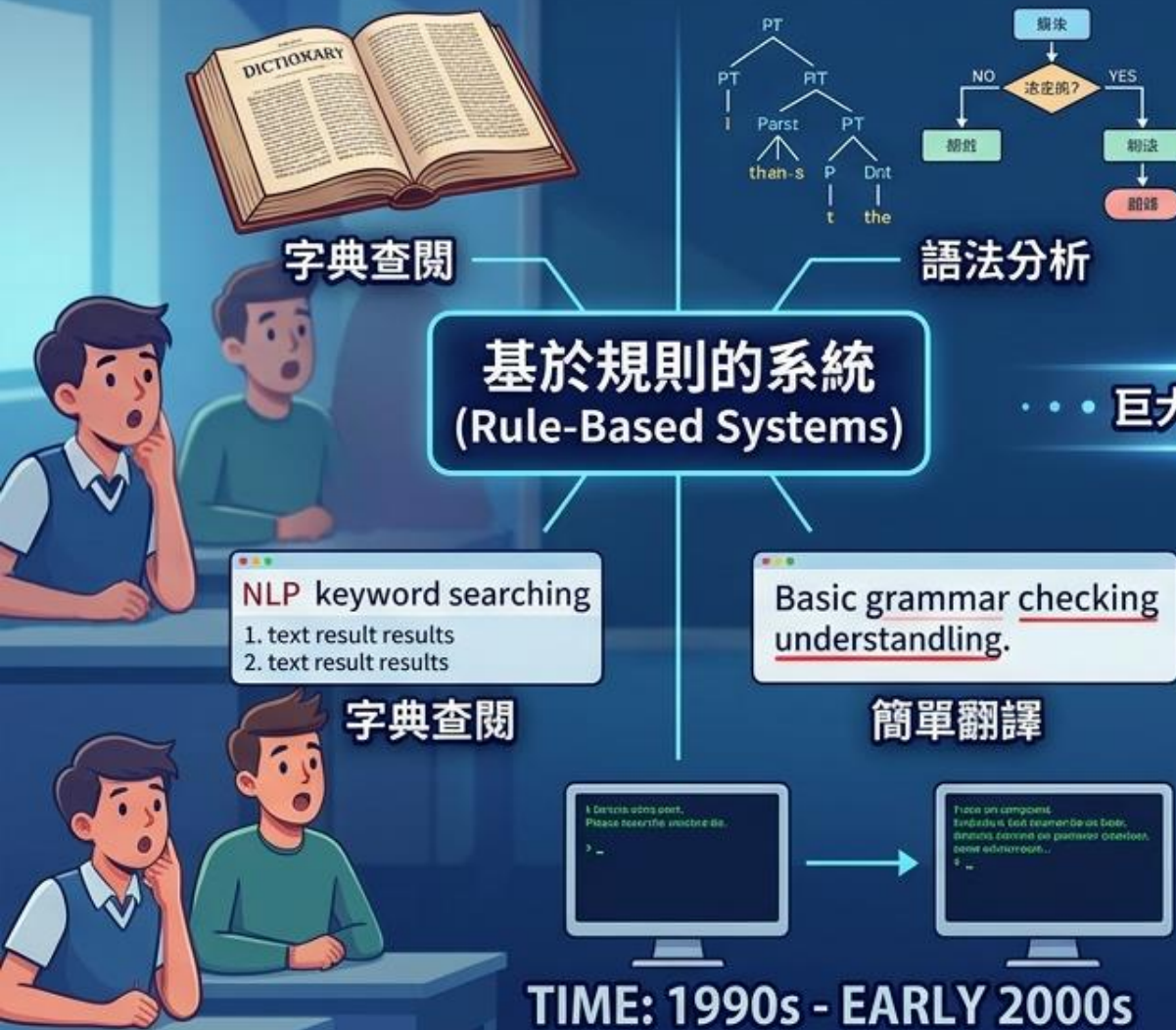
NLP 技術演進與應用差距：從字典到智慧腦

TIME: 1990s - EARLY 2000s

NLP 的過去：規則與基礎

TIME: PRESENT DAY

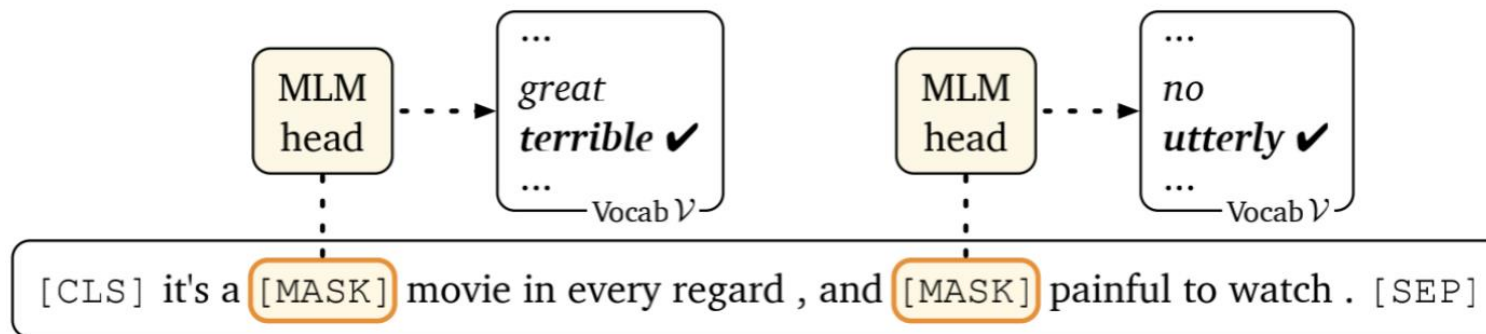
NLP 的現今：深度學習與大語言模型



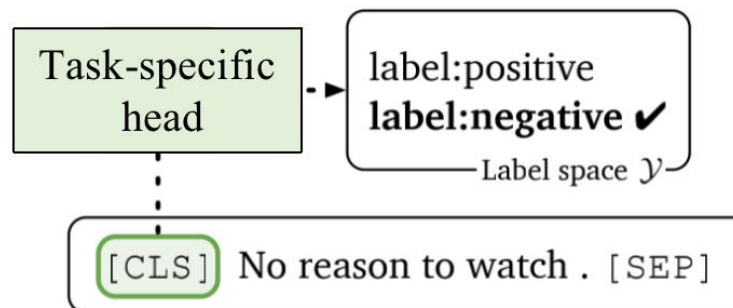
巨大的技術躍遷



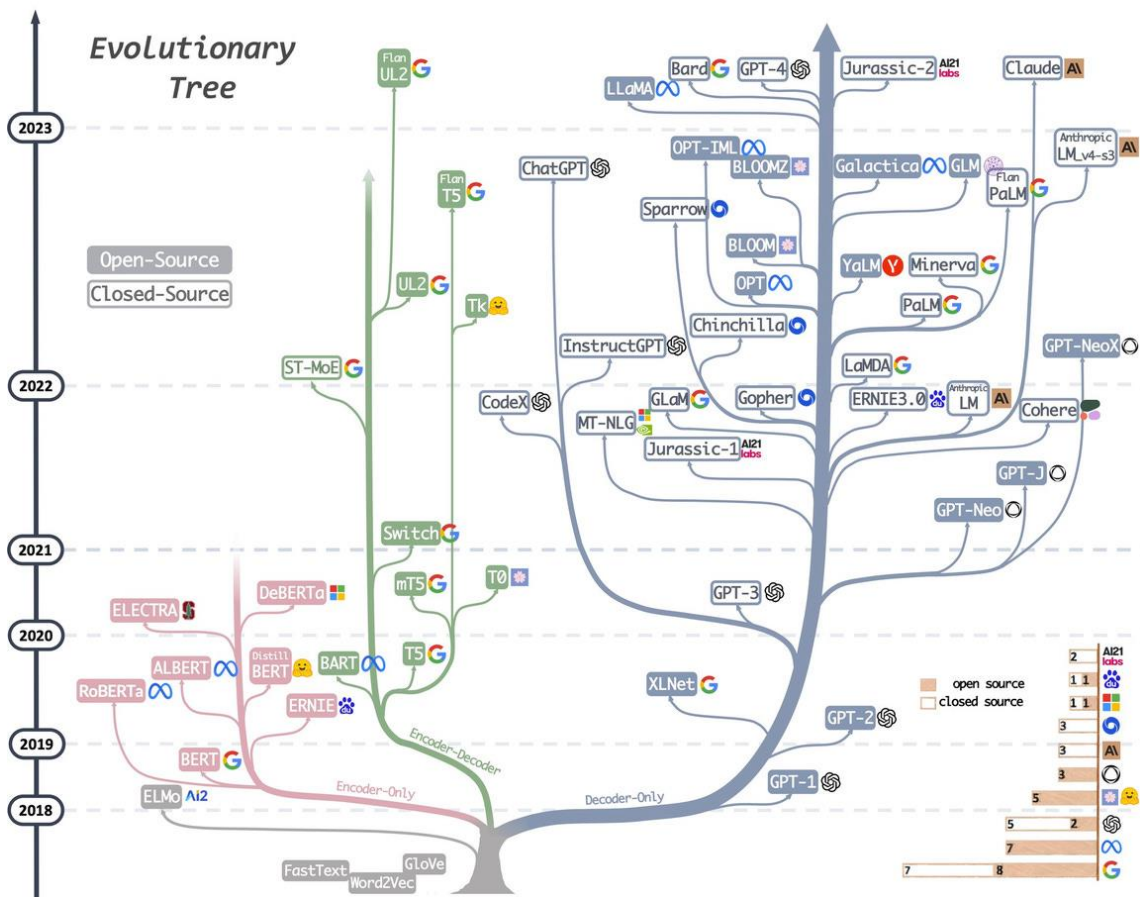
“pre-train (預訓練), fine-tune(微調)” paradigm



(a) MLM pre-training



(b) Fine-tuning



百度 文心一言	阿里巴巴 通义千问	360 360智脑	腾讯 混元	华为 盘古
商汤科技 日日新	澜舟科技 孟子	智源研究院 悟道2.0	科大讯飞 星火认知	昆仑万维 天工3.5
清华大学 ChatGLM-6B	复旦大学 MOSS	中科院自动化所 紫东太初	面壁智能 CPM-Bee	达观数据 曹植

TAIDE
TRUSTWORTHY AI DIALOGUE ENGINE

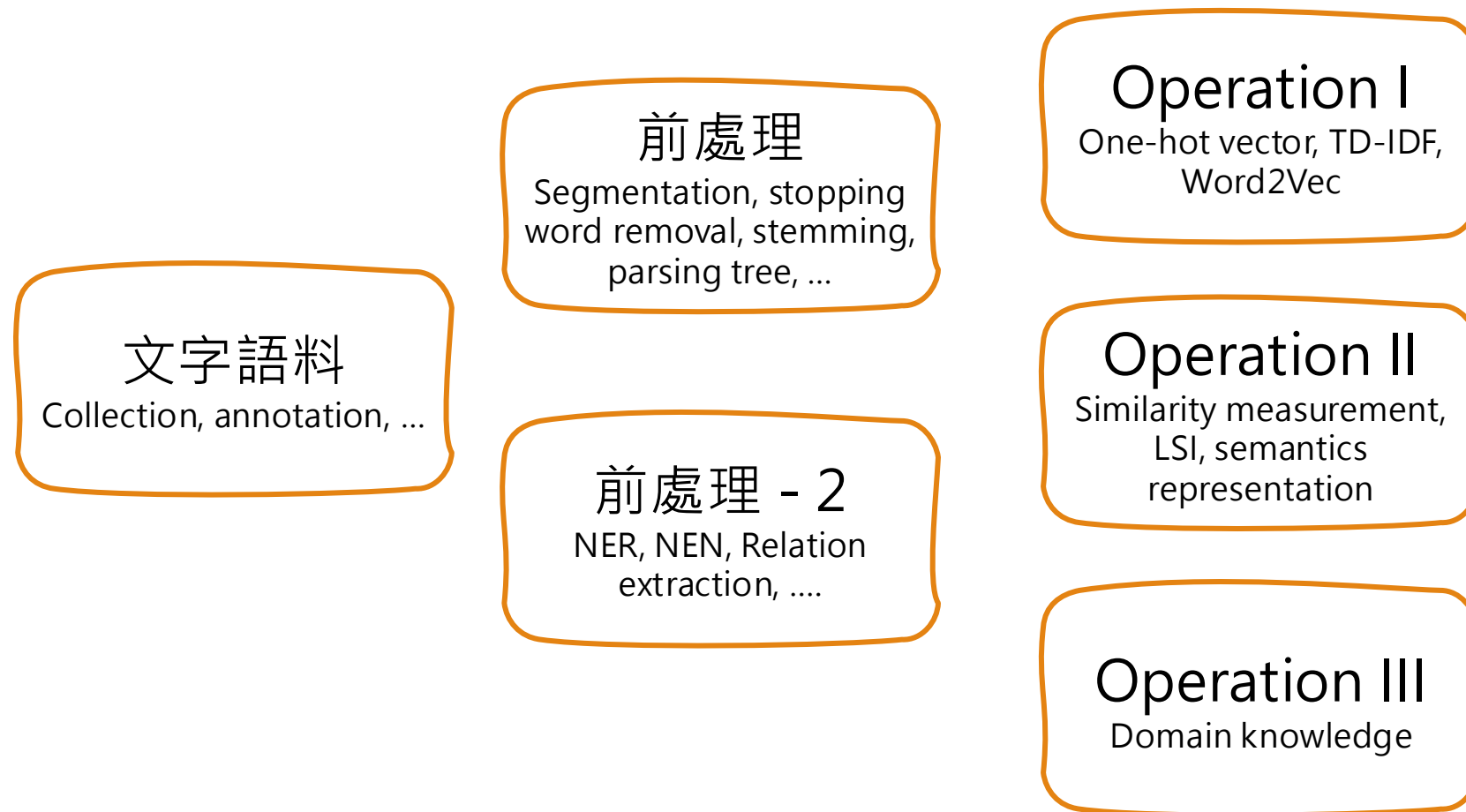


<https://github.com/Mooler0410/LLMsPracticalGuide>

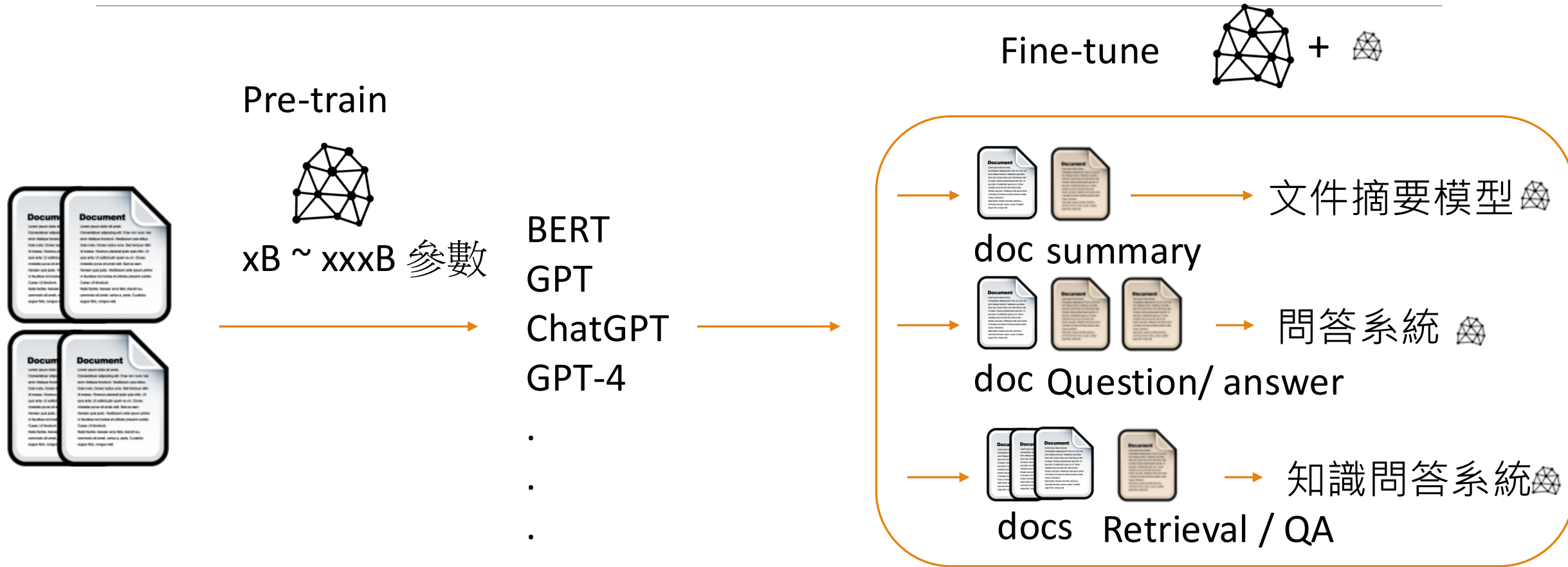
LLM Family Tree



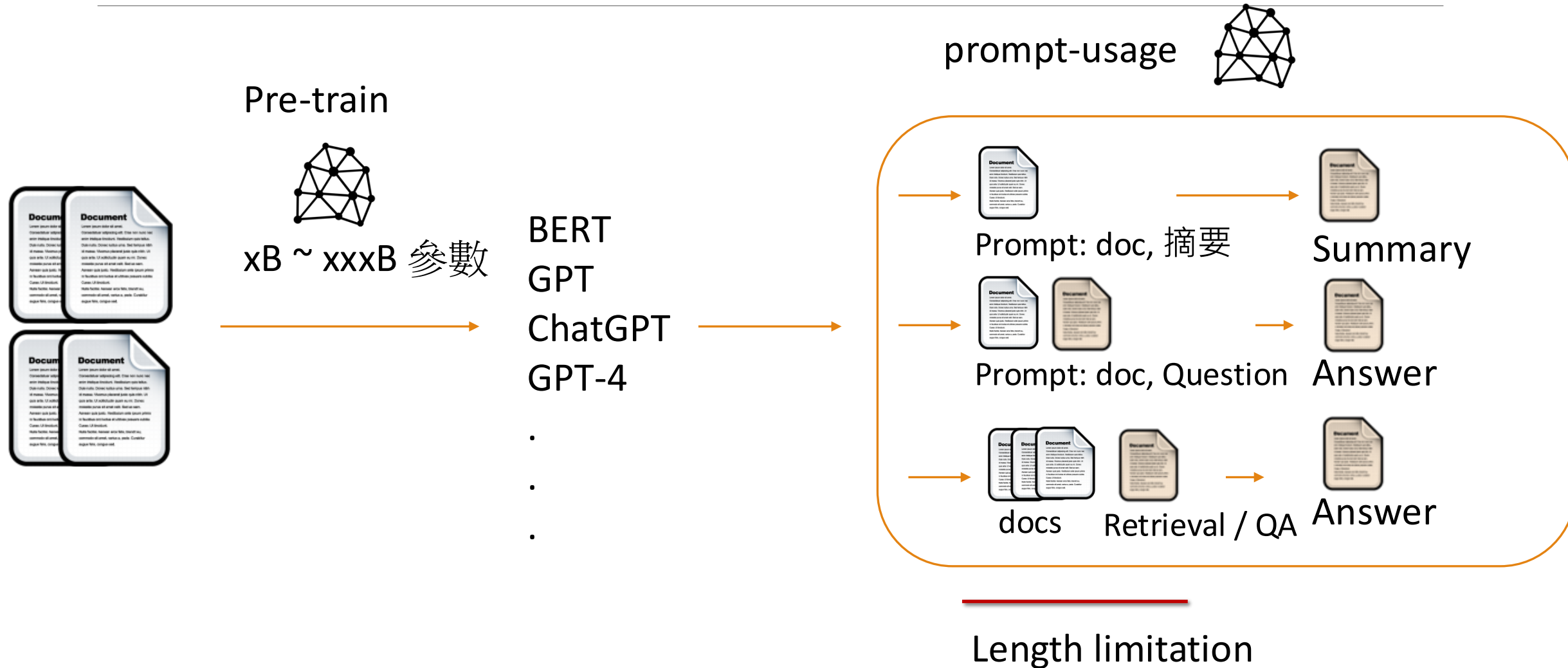
Traditional NLP applications (text)



Large Language Model (LLM) enabled NLP applications



Large Language Model (LLM) enabled NLP applications (no fine-tune)



自然語言處理 基本概念與方法

將介紹NLP的核心概念，包括字詞表示、文字處理、字詞向量表示、語意分析和情感分析等。

1 詞彙分析

分析文字結構，將句子拆解為詞彙並標記詞性。

2 語法分析

解析句子的文法結構，理解詞彙之間的關係。

3 語意理解

理解文字的含義，並分析語句之間的邏輯關係。

4 情感分析

識別文字的情緒和主觀性，了解作者的立場和態度。





深度學習與 大型預訓練語言模型的應用

將探討大型預訓練語言模型 (LLMs) 如何革新NLP領域，並展示其在各項應用中的優勢。

例如BERT、GPT-3等，解決不同語言處理任務。

1

Transformer

重要基礎模型原理。

BERT:雙向編碼器表示模型，善於理解上下文資料。

2

序列類神經網路

RNN, LSTM, Seq2seq。

3

生成式AI

T5
GPT

4

大型語言模型系統架構流程與應用

預訓練
微調
PEFT
提示學習 / In-context learning

In this course

Basics

- Python usage

NLP Basics

NLP with DL

- Sequential model
- Generative Models (text)

NLP issues in LLM era

- Reasoning
- Agentic AI

Not included

- Speech
- Prompt usage
- Develop new models
- Solve GAI problems

Schedule

Lecture

Tutorial

Reporting

Exam

<i>Week</i>	<i>Topics</i>	<i>Note</i>
W1	課程簡介 Syllabus / Introduction to NLP	
W2	自然語言處理簡介 (1/2) Introduction to NLP (vector space, indexing, parts of speech, phrase structure)	
W3	自然語言處理簡介 (2/2) Introduction to NLP (Language model)	
W4	基礎文字資料機器學習 (1/2) Basic machine learning for text (Text Classification, NB, NN)	
W5	基礎文字資料機器學習 (2/2) Basic machine learning for text (word embedding, text representation)	
W6	Python for text tutorial (1/2)	
W7	Python for text tutorial (2/2)	
W8	文字生成式AI簡介(1/3) Introduction to GAI (text): Word Embeddings, Language Modeling (RNN), Sequence-to-sequence Models, and Attention Mechanisms, Sub-word Tokenization; Transformers	
W9	文字生成式AI簡介(2/3) Introduction to GAI (text): ELMo, BERT, GPT, and T5 (BERT and its Family)	
W10	文字生成式AI簡介(3/3) Introduction to GAI (text): Decoding Strategies and Evaluations for Natural Language Generation	
W11	大語言模型簡介與訓練 (1/3): Large language model concept and training (GPT-3, InstructGPT, RLHF)	
W12	大語言模型簡介與訓練 (2/3): Parameter Efficient Fine-Tuning (PEFT)	
W13	大語言模型簡介與訓練 (2/3): Introduction and Review technique of Retrieval Augmented Generation (RAG) I	
W14	Exam	
W15	大語言模型簡介與訓練 (2/3): Introduction and Review technique of Retrieval Augmented Generation (RAG) II	
W16	Reasoning / Agent	

The course content is subject to minor adjustments.



Grading

This course is X-class in NTHU, you can enroll other courses with the same time slots in NTHU.

The exam will be held **in person**.

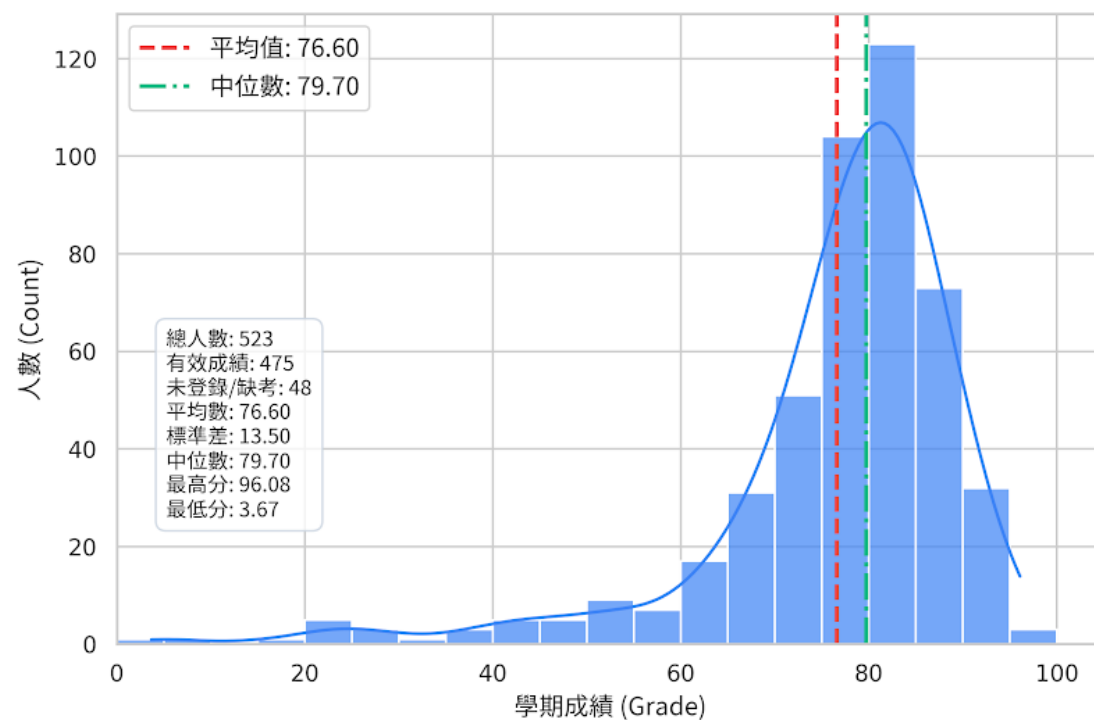
Assignments **75 %**

- **4** assignments for each student
 - Coding needed
 - The loading of Single assignments will be "fine-tuned" according to the alternative designs. Their credits will also be slightly changed (-3%~+3%)

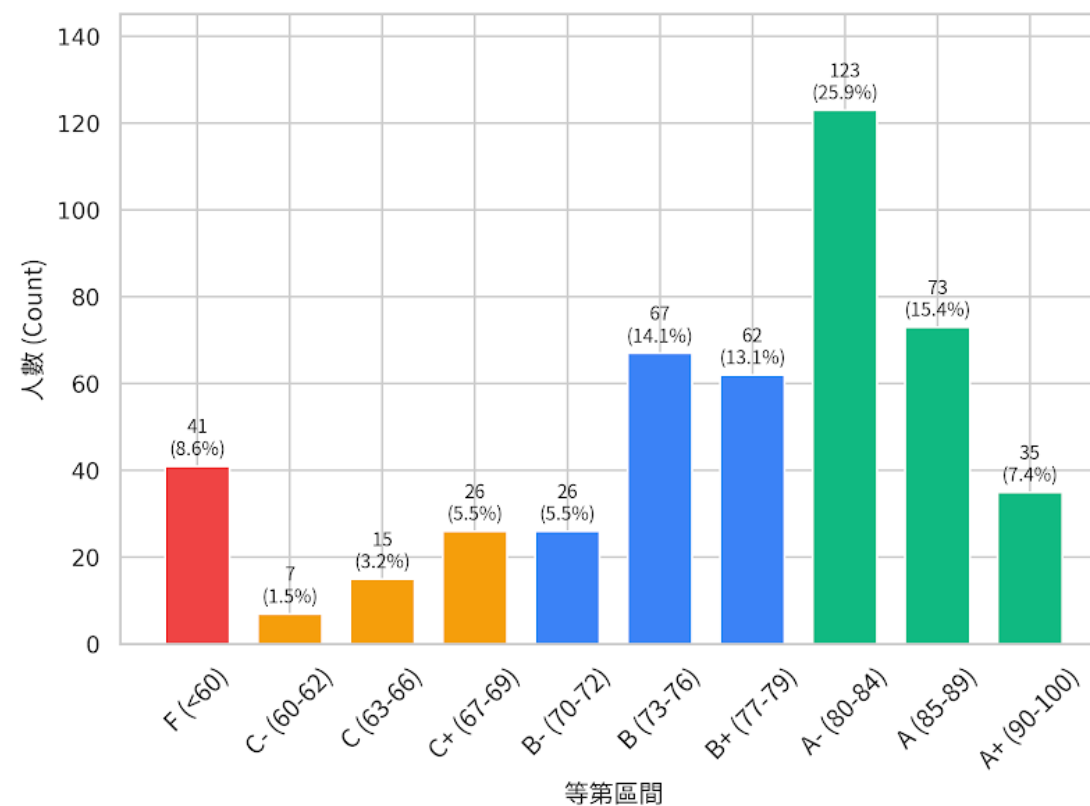
Midterm Exam **25 %** (week 14, Tentative date)

Grade in 2025

學期成績數值分佈 (5分區間)



大學等第成績分佈 (Letter Grade)



Homework evaluation

Human-TA批改

AI-TA 協助

Code / Results
Evaluation

Discussion
Evaluation

Your Insight, not GPT insight

This course

Is designed for

- 對自然語言處理有高度興趣的學生
- 有一些深度學習模型訓練基礎的學生
- 想了解why LLM so power?

The hardships of learning

- 這堂課真的會花很多時間，尤其是學期初的時候，不過在經過多次的練習後，會越來越得心應手
- 因為可以採用非同步上課，所以常常在很認真上課、沒有翹課的情況下不小心遺忘了這堂課的進度，有時候遇到期中考或者是報告跟 NLP 作業重疊在一起的時候真的蠻累的，建議大家好好安排時間

總的來說如果你是對 NLP 有興趣的碩士生且有一定的 coding 跟 DL 基礎的話我會推薦這門課，但是對於有分數跟其他科目壓力的大三以下的學生來說就需要好好評估自己的時間跟能力還有成績，因為我目前看起來這學期的成績分布應該不會如 dcard 那篇心得裡那樣的好看。

我也很喜歡課程後半段有帶到 Hugging Face、LangChain、Ollama 等工具的初步串接，讓我也能了解模型如何和 retrieval 或應用流程結合，這部分非常有趣。若未來能再多補一些這類系統層面的說明我想對學生會更有幫助。

在課程安排的部分，我非常感謝老師安排了很豐富且實用的技術點，並在最後很巧妙的整合、串連成一個連貫的過程。在課程初期的 Word Embedding 中，我學習到如何將文字符號映射到連續的向量空間，讓我感覺這好像是我缺失的拼圖重要的一塊，並且在作業一的實作中，實際操作語意關聯性的計算方法。課程中還有分析過去使用 RNN 常遇到的問題（同時也是我在修 ML 時作業遇到的瓶頸），像是長序列遺忘、無法平行運算等等，讓我學會 Transformer 架構中，透過使用 Self-Attention 機制，可以從根本上改變資訊聚合的方式，讓我很深刻且具體的體會到 LLM 的強大之處。

此外，在學期中後段的課程裡，我學到了參數高效微調 PEFT 的技術，正好對我目前進行的研究有很大的幫助。因為實驗室的 GPU VRAM 資源有限，讓我無法對大型的預訓練模型進行全參數微調，當時也是自己看了許多篇研究論文，剛好結合課堂的介紹，我學會可以使用 LoRA 這類的微調機制，大幅降低對硬體資源的限制，有效的解決我在有限算力下進行特定領域模型 Adaption 的問題，讓我在實驗設計上有非常大的突破。

Instructors and TAs

- Instructors:
 - Hung-Yu Kao 台達館 633 (hykao@cs.nthu.edu.tw)
- TAs:
 - 台達館 7F, Room 714 IKM lab.
- Course website
 - NTU Cool (<https://cool.ntu.edu.tw/courses/41436>)
 - Github (https://github.com/IKMLab/NTHU_Natural_Language_Processing)